

Kernel errors/warnings summary with full severity levels and colorized output

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Retrieves all kernel messages from the kernel ring buffer, extracts and counts all message severity levels (emerg, alert, crit, err, warn, notice, info, debug), and presents them with a comprehensive summary frontmatter followed by the human-readable colorized logs.

Language

bash

Category

diagnostics

Command

```
dmesg -T -x --level=emerg,alert,crit,err,warn --color=always
2>/dev/null | awk 'BEGIN { print "\033[1m=== KERNEL
MESSAGE SUMMARY ===\033[0m" } { lines[NR] = \033[0m; lvl = \
$2; gsub(/\x1B\[0-9;]*[mK]/, "", lvl); gsub(/[: \t]/,
"\n", lvl); if(lvl ~ /^(emerg|alert|crit|err|warn|notice|
info|debug)\$/ ) cnt[lvl]++; if(lvl ~ /^w+\$/ ) all_cnt[
lvl]++ } END { if(NR>0) { print "\033[1m-- ERRORS &
WARNINGS ---\033[0m"; for(l in cnt) printf "\t %-7s : %d\n",
l, cnt[l]; print "\033[1m--- ALL MESSAGES (by
severity) ---\033[0m"; for(l in all_cnt) printf "\t %-7s
: %d\n", l, all_cnt[l]; print "\033[1m
===== \033[0m\n"; for(i
=1; i<=NR; i++) print lines[i] } else print "No kernel
messages found for selected levels." }'
```

Explanation

The one-liner uses 'dmesg -T -x' to extract kernel messages with human-readable timestamps (-T) and decode the facility/level prefixes (-x). We filter for all severity levels (emerg,alert,crit,err,warn,notice,info,debug) via '-level'. The '-color=always' flag preserves standard color coding. The output is piped into 'awk', which does several tasks: (1) strips ANSI escape sequences from the second column to cleanly count occurrences, (2) maintains two counters: one for 'errors/warnings' (emerg,alert,crit,err,warn) and another for 'all messages'

(all severity levels), and (3) stores all log lines in memory. The END block prints a formatted summary frontmatter with both the 'errors/warnings' and 'all messages' counts, followed by the original colorized logs.

Tags

dmesg, kernel, errors, warnings, logs, troubleshooting, monitoring, summary, awk, diagnostics, severity, log-analysis, linux, system, administration

Dependencies

util-linux, gawk

Arguments

None

Examples

1. `dmesg -T -x --level=emerg,alert,crit,err,warn --color=always 2>/dev/null`
`| awk 'BEGIN { print "\033[1m=== KERNEL MESSAGE SUMMARY ===\033[0m"`
`} { lines[NR] = \033[0; lvl = \033[2; gsub(/\033[x1B\033[[0-9;]*[mK]/,`
`\033["\033", lvl); gsub(/[: \033t]/, \033"\033", lvl); if(lvl ~ /^(emerg|alert|crit|err|warn`
`cnt[lvl]++; if(lvl ~ /\033w+\033$/) all_cnt[lvl]++ } END { if(NR>0)`
`{ print "\033[1m--- ERRORS & WARNINGS ---\033[0m"; for(l in`
`cnt) printf \033" %-7s : %d\033n\033", l, cnt[l]; print "\033[1m---`
`ALL MESSAGES (by severity) ---\033[0m"; for(l in all_cnt) printf`
`\033" %-7s : %d\033n\033", l, all_cnt[l]; print "\033[1m=====`
`for(i=1; i<=NR; i++) print lines[i] } else print "No kernel messages`
`found for selected levels." }' - Show full severity summary and col-`
`orized kernel messages`
2. `dmesg -T -x --level=emerg,alert,crit,err,warn --color=always 2>/dev/null`
`| awk 'BEGIN { print "\033[1m=== KERNEL MESSAGE SUMMARY ===\033[0m"`
`} { lines[NR] = \033[0; lvl = \033[2; gsub(/\033[x1B\033[[0-9;]*[mK]/,`
`\033["\033", lvl); gsub(/[: \033t]/, \033"\033", lvl); if(lvl ~ /^(emerg|alert|crit|err|warn`
`cnt[lvl]++; if(lvl ~ /\033w+\033$/) all_cnt[lvl]++ } END { if(NR>0)`
`{ print "\033[1m--- ERRORS & WARNINGS ---\033[0m"; for(l in`
`cnt) printf \033" %-7s : %d\033n\033", l, cnt[l]; print "\033[1m---`
`ALL MESSAGES (by severity) ---\033[0m"; for(l in all_cnt) printf`
`\033" %-7s : %d\033n\033", l, all_cnt[l]; print "\033[1m=====`
`for(i=1; i<=NR; i++) print lines[i] } else print "No kernel messages`
`found for selected levels." }' | less -R - Show the summary and`
`logs with a pager, preserving colors`

```

3. sudo dmesg -T -x --level=emerg,alert,crit,err,warn --color=always
2>/dev/null | awk 'BEGIN { print "\{\}033[1m=== KERNEL MESSAGE SUMMARY
===\{\}033[0m" } { lines[NR] = \{\}$0; lvl = \{\}$2; gsub(/\{\}x1B\{\}[[0-9;]*[mK]/,
\{\}"\{\}", lvl); gsub(/[: \{\}t]/, \{\}"\{\}", lvl); if(lvl ~ /^(emerg|alert|crit|err|warn
cnt[lvl]++; if(lvl ~ /\{\}w+\{\}$/) all_cnt[lvl]++ } END { if(NR>0)
{ print "\{\}033[1m--- ERRORS & WARNINGS ---\{\}033[0m"; for(l in
cnt) printf \{\}" %-7s : %d\{\}n\{\}", l, cnt[l]; print "\{\}033[1m---
ALL MESSAGES (by severity) ---\{\}033[0m"; for(l in all_cnt) printf
\{\}" %-7s : %d\{\}n\{\}", l, all_cnt[l]; print "\{\}033[1m=====
for(i=1; i<=NR; i++) print lines[i] } else print "No kernel messages
found for selected levels." }' - Run with sudo to access kernel ring
buffer on restricted systems

```

Output

```

\u001b[1m=== KERNEL MESSAGE SUMMARY ===\u001b[0m
\u001b[1m--- ERRORS & WARNINGS ---\u001b[0m
    err      : 2
    warn     : 3
    crit     : 1
\u001b[1m--- ALL MESSAGES (by severity) ---\u001b[0m
    err      : 2
    warn     : 3
    crit     : 1
    notice   : 5
    info     : 12
    debug    : 0
\u001b[1m=====\u001b[0m

kern :err   : [Tue Apr 21 10:00:00 2026] sd 0:0:0:0: [sda]
      Write cache: enabled
kern :warn  : [Tue Apr 21 10:01:23 2026] WARNING: CPU: 0
      PID: 1234
kern :crit  : [Tue Apr 21 10:02:45 2026] Critical hardware
      error
kern :warn  : [Tue Apr 21 10:05:00 2026] ext4-fs: mounted
      with warnings
kern :err   : [Tue Apr 21 10:06:12 2026] usb 1-1: device
      descriptor read/64, error -71
kern :warn  : [Tue Apr 21 10:08:34 2026] IPv6: ADDRCONF(
      NETDEV_UP): eth0: link is not ready
kern :notice: [Tue Apr 21 10:10:00 2026] Kernel log system
      initialized
kern :info  : [Tue Apr 21 10:15:00 2026] Network interface
      eth0 up and running

```

Notes

- Requires read access to the kernel ring buffer, which often requires sudo or being in the 'adm' group on some distributions.
- The '-x' (-decode) flag is critical here as it explicitly outputs the severity level on every line, making awk parsing reliable.
- ANSI stripping regex in awk ensures color codes don't interfere with severity counting.
- This command counts all message severity levels: emerg, alert, crit, err, warn, notice, info, debug.
- The 'errors & warnings' summary includes: emerg, alert, crit, err, warn.
- The 'all messages' summary includes all severity levels, providing complete log statistics.
- The entire dmesg output is stored in awk's memory (the 'lines' array) before printing, which is perfectly safe for typical volumes but could consume slightly more memory on massively flooded systems.

Warnings

- Running without sufficient privileges (like sudo) will result in a blank output or a permission denied error on most modern secure Linux distributions.

See Also

- find-large-files-recursive
- disk-space-sort-largest-directories

Status

reviewed

Safety

safe

Shell

bash

Platforms

linux

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